Credit-card accounts

ProfitSync · engineering report · branch `feat/credit-card-accounts` · 4 September 2026

A credit card is now a first-class **liability** account. Buying with the card records an expense and increases the card's debt; paying the card is a transfer from a bank or cash account that reduces the debt and is never a second expense. Statements are tracked per billing cycle, with partial payments, refunds, fees, overpayments and full trash/restore/edit reversibility. Available credit is shown but never counted as money the user owns.

735	**10 / 10**	**1**	**1**
unit tests green (64 files, +64 new)	end-to-end scenarios on a real database	migration (0062), additive, legacy rows untouched	pre-existing money bug found and fixed (Trash restore)

1. What the ledger looked like before

- Accounts (`wealth_accounts`) had `type ∈ bank | cash | space` and a stored `current_balance` that ten code paths mutate with relative SQL deltas. The sign lives in one place: `balanceDelta(type, amount)` in `src/lib/wealth-ledger.ts` (+incoming / −outgoing), reused on create, edit, trash, restore and purge.
- Transactions: `type ∈ incoming | outgoing` , positive amounts, `kind ∈ standard | transfer` . A transfer is two legs sharing a `group_id` . `is_system` marks Opening Balance / Balance Adjustment rows, which define a balance and are never reversed through Trash.
- Every report (analytics, money flow, calendar, transactions summary, client totals, budgets v1/v2) inlined its own `case when type = 'incoming'` SQL. There was no refund concept outside Budget v2's provisional "same-category inflow" guess.

2. Architecture chosen — one sign convention

> `current_balance` **stays the signed asset-equivalent value for every account type.** A credit card's balance is therefore normally negative: −950 means "€950 owed". Only presentation reads the sign, through `src/lib/credit-card.ts` (`cardDebt` , `cardCredit` , `availableCredit` , `creditUsage` , `signedBalanceFromDebt`). No component writes a minus sign.

Because of that single rule the existing ledger needed no special cases:

Event	Ledger rows	Card	Bank	Expense	Income	Net worth
Buy €100 groceries with Visa	outgoing €100 on Visa	−100	—	+100	0	−100
Pay €100 Intesa → Visa	transfer (out on Intesa, in on Visa)	+100	−100	0	0	0
Refund €100 to Visa	**refund** (incoming, `kind='refund'`)	+100	—	−100	0	+100
€10 annual fee	outgoing €10 on Visa	−10	—	+10	0	−10
Overpay €150 on €100 debt	transfer	+50 (card credit)	−150	0	0	0

Refunds are a first-class kind

`transactions.kind` gains `refund`: an incoming row that gives money back for an earlier expense. Its balance effect is that of any incoming; reporting nets it against **expense**, never income. The rule exists once in JS (`src/lib/tx-classify.ts`) and once in SQL (`api/_lib/tx-sql.ts`). Every aggregate route imports the SQL twins, and a convention test fails if any route re-inlines the old expression.

Statements: an immutable snapshot table, derived payments

`credit_card_statements` holds one row per closed cycle (`closing_date`, `due_date`, `statement_balance`, `source ∈ computed | manual`). Rows are filed lazily and idempotently when the card is read after a closing date has passed, or entered by the user when onboarding a card with a known statement.

Payments are never stored. Remaining = `clamp(statement_balance − Σ incoming transfer legs dated after the close, 0, statement_balance)`. A later statement already includes older unpaid debt, so one payment reduces every statement it is dated after and the oldest reaches zero first — FIFO with no allocation table that could drift. Trash, restore and edit of a payment recompute deterministically. Post-close refunds are not payments (they post to the current cycle, like a real issuer). New-cycle spending is a separate, gross, historical figure; paying the card never "un-spends" it.

The balance at a close is reconstructed from the authoritative stored balance: `debtAtClose = −(current_balance − movementAfterClose)`, using the same "effect currently applied" rule Budget v2 uses.

Dates

Fixed day-of-month settings (1–31) clamped at use: closing day 31 → Feb 28/29, Apr 30. The due date is the first matching day strictly after the close. UTC ISO strings throughout (same conventions as `recurring.ts`). Tested for 28/29/30/31, leap years including 2000 and 2100, and the year boundary.

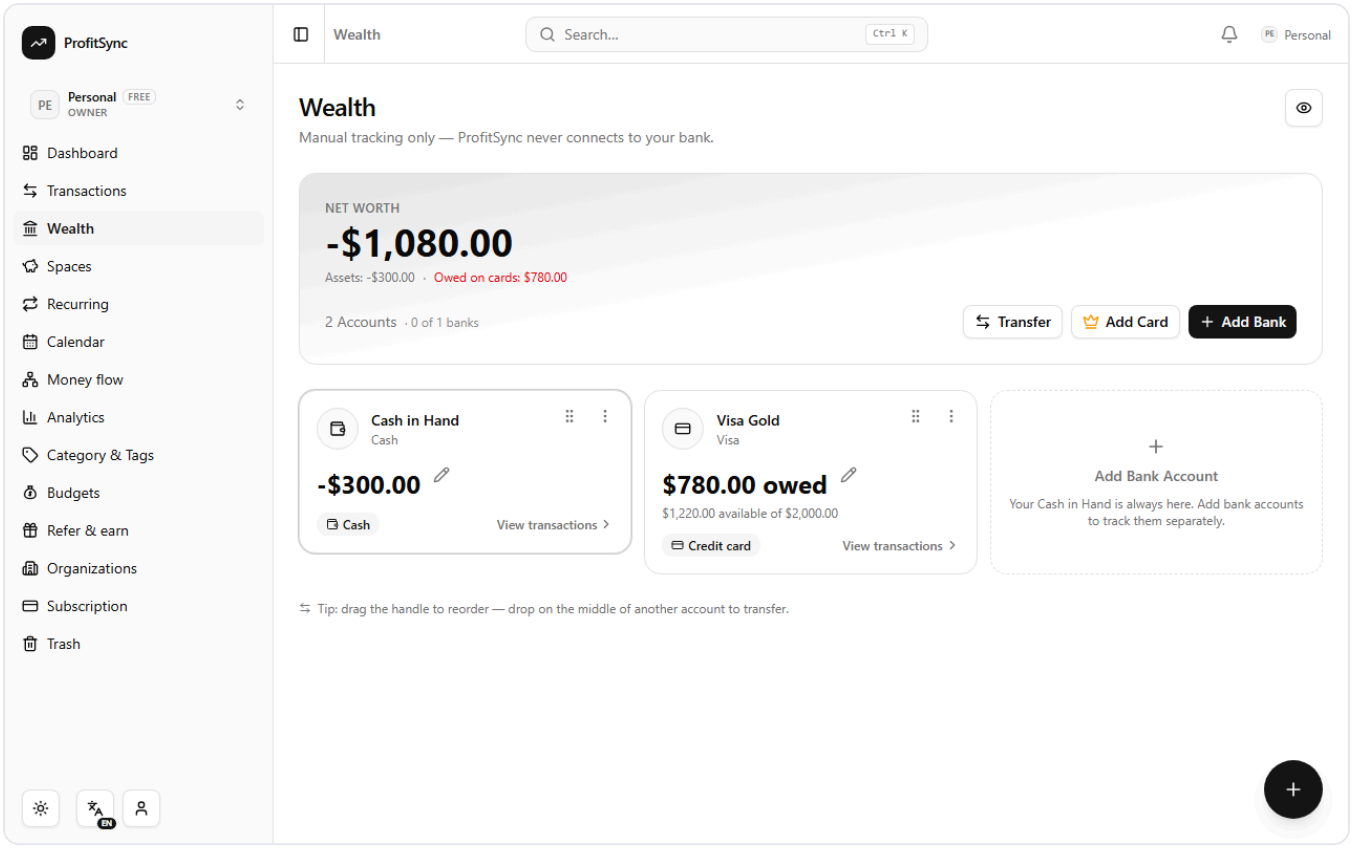
3. Data model

Change	Detail
`wealth_accounts`	`type='credit_card'`; new nullable `credit_limit`, `statement_closing_day`, `payment_due_day` (CHECK 1..31). Existing bank/cash/space rows keep NULLs and are never reinterpreted.
`credit_card_statements` (new)	FK cascade to account and org (factory reset and org deletion leave nothing behind); unique `(wealth_account_id, closing_date)` is the filing idempotency key.
`transactions.kind`	Accepts `refund` (text column, no DDL).
Migration	`drizzle/0062_credit_card_accounts.sql` + journal entry. Applied to the local database and verified in `information_schema`. 0060–0062 are hand-written; `drizzle-kit generate` is out of sync with them (documented in CLAUDE.md).

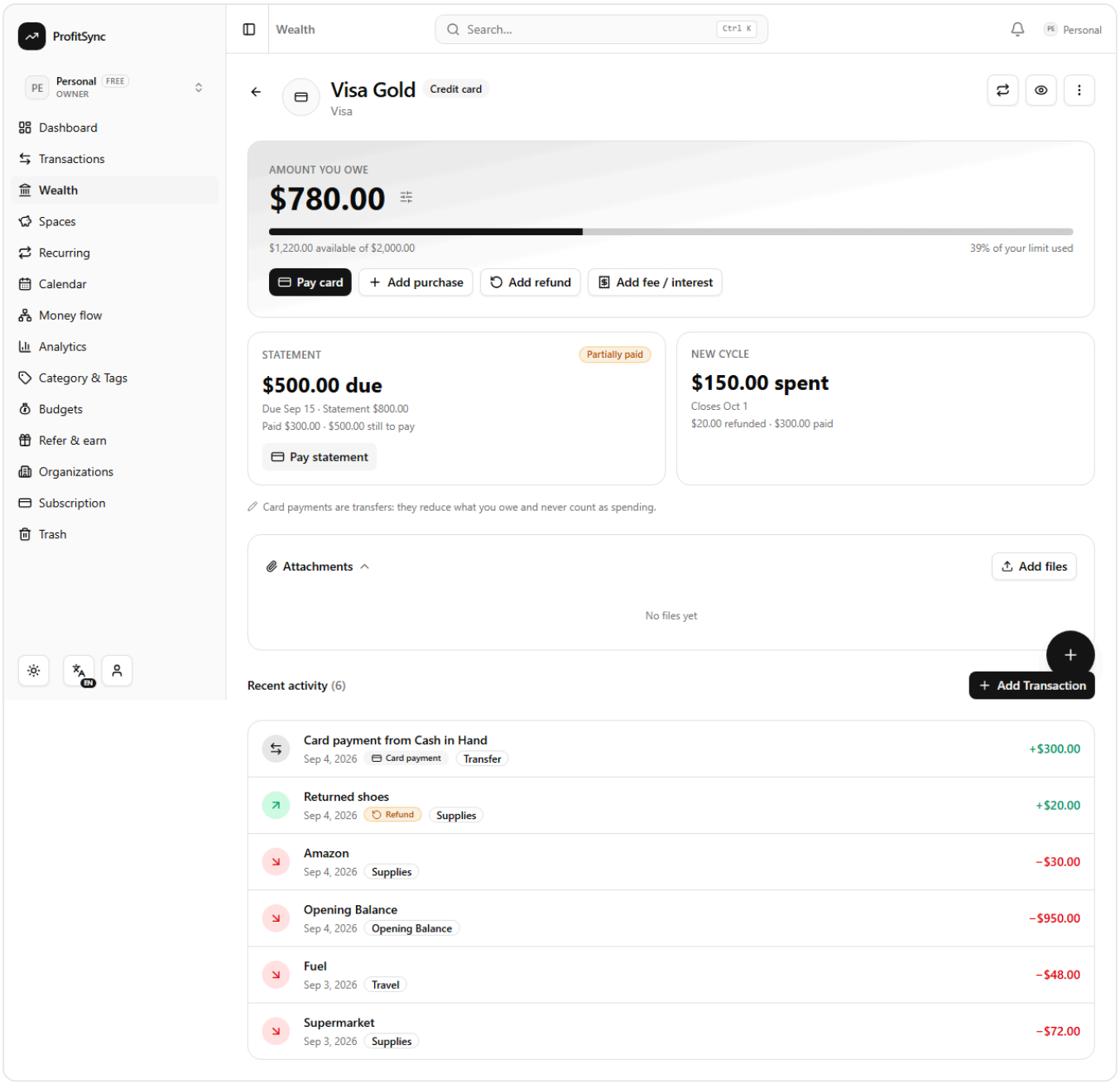
4. API

Endpoint	Behaviour
`POST /api/wealth/accounts`	`type: "credit_card"` with `credit_limit`, `current_debt`, `statement_closing_day`, `payment_due_day`, optional `statement { balance, closing_date, due_date? }`. Opening debt becomes a negative opening balance plus an outgoing Opening Balance system row (no expense, no budget spend). A known statement seeds a `manual` statement.
`PATCH /api/wealth/accounts/:id`	Accepts `current_debt` (converted to the signed balance server-side), `credit_limit`, closing/due days.
`GET /api/wealth/accounts/:id/card`	Owed / credit / available / utilisation, latest statement (paid, remaining, textual status), older statements, open cycle (spent, refunds, payments, closes on, next due).
`POST /api/wealth/transfer`	Paying a card is this endpoint with the card as `to_account_id`; legs are labelled "Card payment to / from …".
`POST /api/transactions`, `/group`, `PATCH /:id`	Accept `kind: "refund"` (must be incoming). A transfer leg refuses kind/direction/account edits.
Quota	New `creditCards` plan limit (free: 1 active, paid: 20 incl. closed), exposed by `GET /api/wealth/quota`.

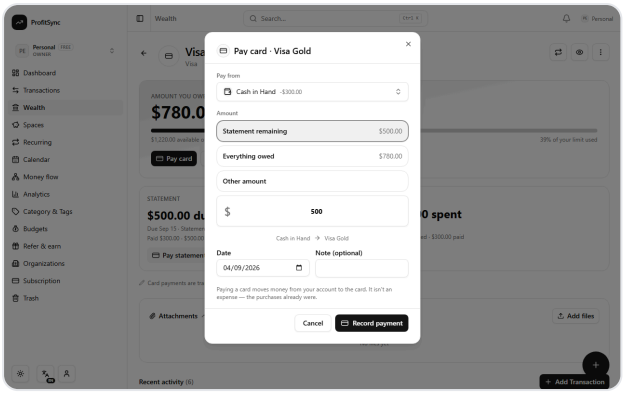
5. The built UI

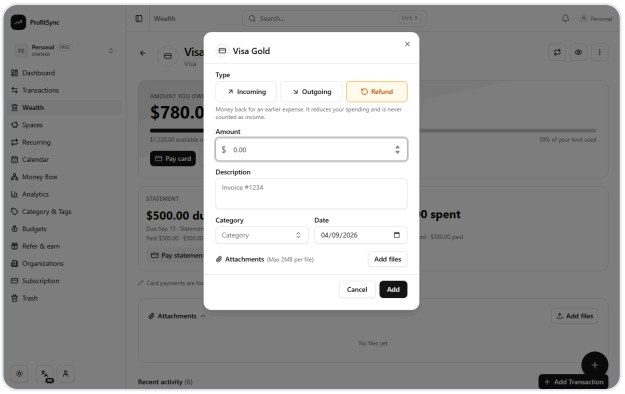


/wealth — the card tile says what is owed and what is available; net worth shows Assets and Owed on cards separately. (Demo data: the payment was made from a €0 cash account, hence the negative cash balance.)

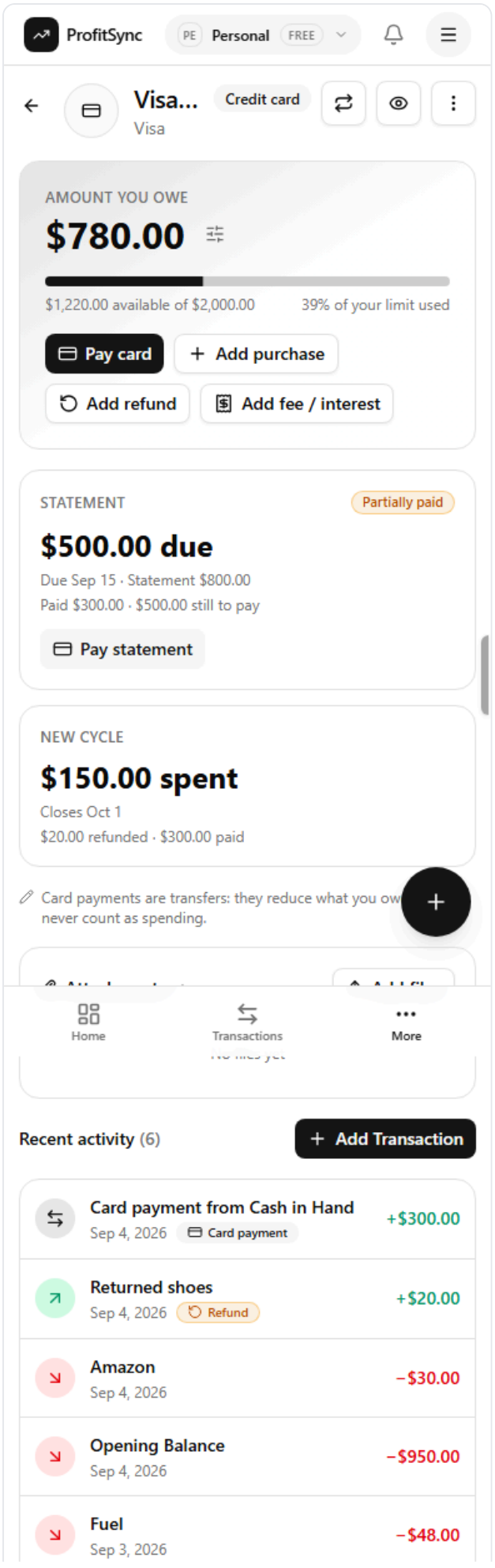


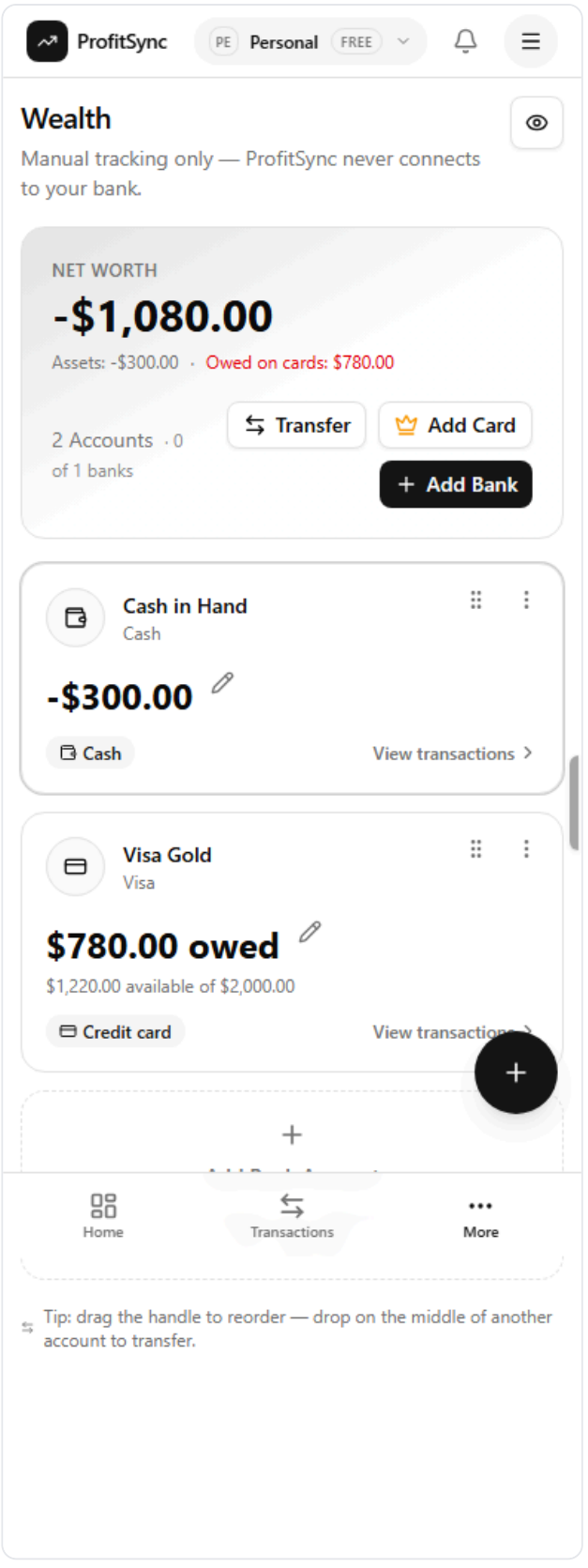
/wealth/:id for a card — amount you owe, utilisation bar with accessible value text, STATEMENT (textual status, paid, still to pay, Pay statement), NEW CYCLE, quick actions, and recent activity with Card payment / Refund badges.

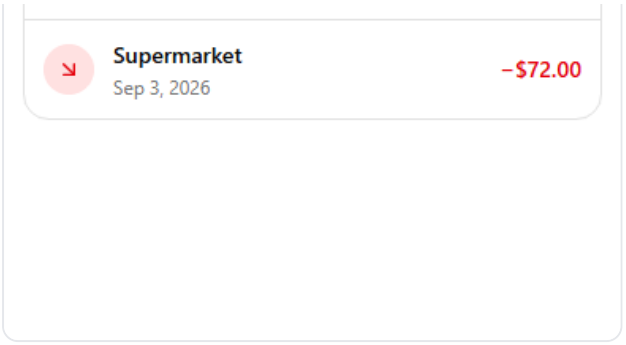




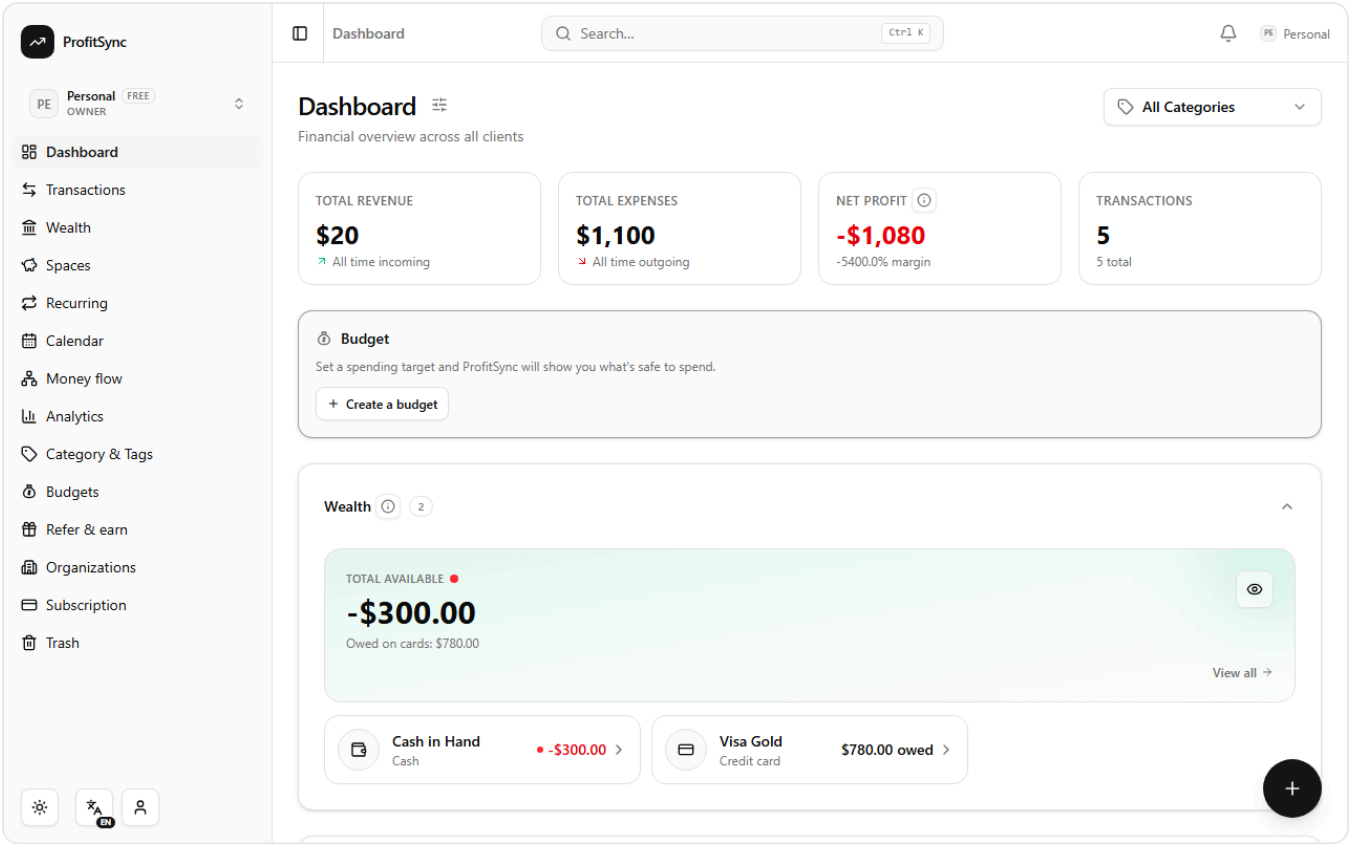
Left: Pay card — statement remaining / everything owed / other amount, from a bank or cash account; recorded as a transfer. Right: the Income / Expense / Refund selector (a refund picks from expense categories).


ProfitSync
PE Personal FREE
Visa...
Credit card
Visa
AMOUNT YOU OWE
$780.00
$1,220.00 available of $2,000.00
39% of your limit used
Pay card
Add purchase
Add refund
Add fee / interest
STATEMENT
Partially paid
$500.00 due
Due Sep 15 · Statement $800.00
Paid $300.00 · $500.00 still to pay
Pay statement
NEW CYCLE
$150.00 spent
Closes Oct 1
$20.00 refunded · $300.00 paid
Card payments are transfers: they reduce what you ow
never count as spending.
Home
Transactions
More
Recent activity (6)
Add Transaction
Card payment from Cash in Hand
Sep 4, 2026
Card payment
+$300.00
Returned shoes
Sep 4, 2026
Refund
+$20.00
Amazon
Sep 4, 2026
−$30.00
Opening Balance
Sep 4, 2026
−$950.00
Fuel
Sep 3, 2026
−$48.00

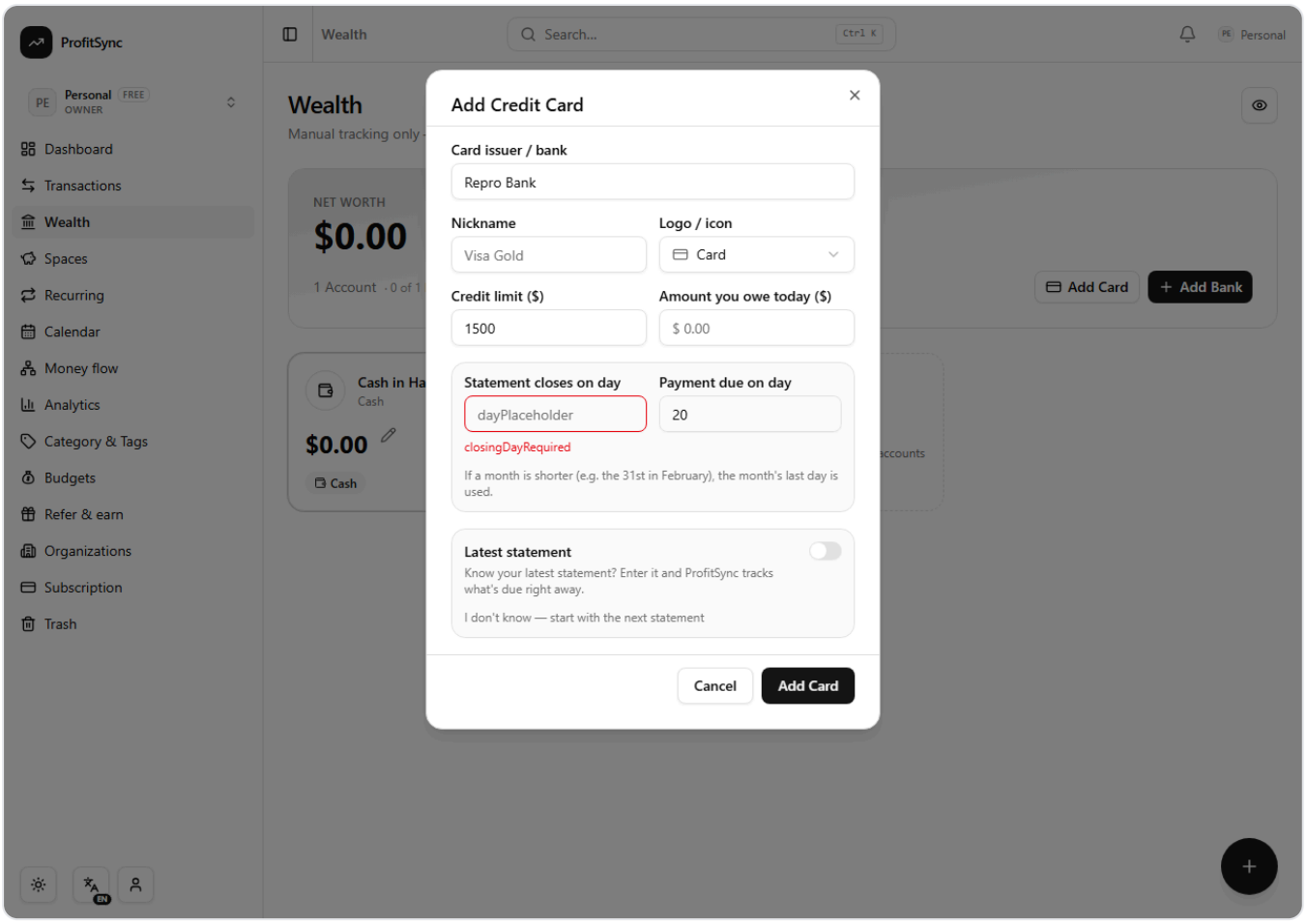

ProfitSync
PE Personal FREE
Wealth
Manual tracking only — ProfitSync never connects to your bank.
NET WORTH
-$1,080.00
Assets: -$300.00 · Owed on cards: $780.00
2 Accounts · 0
of 1 banks
Transfer
Add Card
Add Bank
Cash in Hand
Cash
-$300.00
Cash
View transactions
Visa Gold
Visa
$780.00 owed
$1,220.00 available of $2,000.00
Credit card
View transactions
Home
Transactions
More
Tip: drag the handle to reorder — drop on the middle of another account to transfer.



Mobile (390 px): the same DOM the Capacitor shells run — 44 px targets, no horizontal scroll.



Dashboard — Total available is liquid money only; Owed on cards is listed beneath; a card's negative balance is never flagged as overdrawn.



Add Card — field-level validation (after the first-review fix: an empty day field now names itself instead of showing the generic month-length hint).

6. Integration impact

Area	Effect
Budgets v1	Spend predicates include refunds as negative spend (`budgetSpendSignedAmount`); card payments are transfers and never match.
Budgets v2	A `refund` row counts as a **confirmed** refund in its category's envelope (catch-all included), never as a provisional guess; envelope detail lists refund rows.
Analytics · Money flow · Calendar · Transactions summary · Client totals	All use the shared aggregates: transfers count nowhere, refunds reduce expense, income excludes refunds.
Net worth	Σ signed balances (debt reduces it); a payment leaves it unchanged; available credit is never added.
AI quick add / voice	Schema gains `kind` , `to_account_name` , `statement_payment` ; accounts are labelled by type in the prompt. "Paid 500 towards Visa from Intesa" → a transfer saved through the transfer endpoint. "Paid my Visa statement" → amount filled from the statement remaining and flagged for review. Quick-fill refuses to fill a transfer into the expense form.
Trash	Delete / restore / purge verified for purchases and payments. **Fixed:** restore only brought back the one leg whose id was passed, leaving the other leg of a transfer or split in Trash — restoring a card payment reduced the debt without re-debiting the bank. Restore now brings back every trashed leg of the group.
Persistence / factory reset	Statements cascade with the account; no change to `resetOrgData` or org teardown.
i18n	~100 new keys in `en.json` , propagated to the other 7 locales with the repo's English-placeholder strategy (translations pending, per convention).
Native	Android synced (`cap:sync:android`), iOS bundle built and copied.

7. Tests

Suite	What it pins
`src/lib/credit-card.test.ts` (32)	Sign convention, available credit, day clamping incl. leap years (2000, 2100), year boundary, due-date rule, open cycle, statement/new-cycle boundary, filing schedule, snapshot reconstruction, full / partial / multiple / excess / over payments, refund after close, two unpaid statements FIFO, onboarding with a manual statement, validation.
`src/lib/credit-card-ledger.test.ts` (12)	In-memory ledger using the exact route helpers: purchase, payment, refund, fee, overpayment, delete/restore/purge of purchase and payment, edit 100→70, budget double-count, net worth, onboarding, no cross-account leakage.
`tx-classify.test.ts`, `api/_lib/tx-sql.test.ts`, `budget-spend.test.ts`, `budget-engine-refund.test.ts`	Reporting rules in JS and SQL; convention check that no route re-inlines a type-only sum; Budget v1 predicates; Budget v2 refund netting.
`e2e/credit-card.spec.ts` (9 scenarios)	Through the real UI, auth and database: create a card with a known statement → purchase → partial payment → full payment → refund → fee → overpayment → trash/restore/edit exactly once → reload; the paying account moved by exactly the payments.

Commands and results

```
node scripts/secret-scan.mjs                       exit 0
node scripts/check-esm-extensions.mjs              228 files reachable — OK
node scripts/boot-functions.mjs                    all functions boot under Node ESM
node scripts/check-route-guards.mjs                125 handlers guarded
npm run i18n:check                                 8 locales in sync
npm run lint · npm run typecheck                   clean
npx vitest run                                     64 files, 735 tests passed
npm run build + chunk-cycle grep                   built; no charts-* in vendor
playwright e2e/credit-card.spec.ts (local DB)      10/10 passed
playwright e2e/smoke.spec.ts (regression)          9/9 passed
npm run cap:sync:android · build:ios + cap copy    OK
```

8. Known limitations and follow-ups

- Filed statements cannot yet be edited or deleted from the UI (manual correction UX).
- No minimum-payment, interest or grace-period logic; available credit is linear (limit + balance).
- No per-card currency yet (accounts have no currency column); the math takes plain numbers so a card currency can be added without touching it.
- Non-English locales carry English placeholders for the new keys.
- The account-scoped Income/Expenses summary on bank pages still includes system rows (pre-existing, untouched).

> **Review focus.** (1) The transfer-leg edit guard now returns 400 if a client tries to change a transfer leg's kind, direction or account — existing UI never did that. (2) A purchase back-dated after its cycle was filed

> will not change that statement (by design). (3) The Trash restore fix changes behaviour for splits too: restoring one leg restores the whole group, matching delete.

Full design notes: `docs/credit-cards/CREDIT_CARDS.md`. Handoff prompt for the reviewing agent: `docs/credit-cards/HANDOFF_PROMPT.md`.